Comparison with top load and front load washing machine

Sl. No.	Top load washing machine	Front load washing machine
1.	Low efficiency.	High efficiency.
2.	The washing drum is a simple type.	The washing drum is complicated type of cylinder
3.	Normal soap powder is used.	Special soap powder is used
4.	Only certain machines have a heating element inside.	All machines have a heating element inside.
5.	Cost is low	Cost is high

4.4 Electric pump

Water pump is used to suck water from underground to tank. When the motor is turned on, due to the vacuum created by the centrifugal force, water is sucked out to a required place. A single phase motor is sufficient for domestic level, since 2000 to 5000 litres of water per day is required.

Based on the structure, the various types are given below:

1. Centrifugal pump
2. Jet pump
3. Submersible water pump
4. Air compressor

4.4.1 Centrifugal pump

The centrifugal pump is a simple electric motor appliance as in figure 4.24. Normally, 0.5 to 3 horse power single phase capacitor start induction motor is used in centrifugal pump. By using centrifugal force, water is sucked and delivered. This is called a centrifugal pump.

Parts of a centrifugal pump

i. Base plate
ii. Water pump box
iii. Impeller
iv. Shaft
v. Rope and box
vi. Bearings

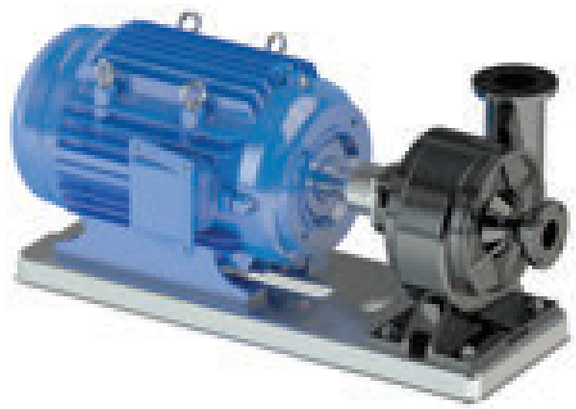


Fig. 4.24 **Centrifugal pump**

i. Basement plate

The basement plate is made up of cast iron or hard steel metal. It is fitted over the base plate with bolts and nuts.

ii. Water pump box

This is usually made up of close-grained cast iron. The vertical plane at the centre of the casing is split into two halves with flanges tightened together by bolts and nuts with gasket for leak proof.

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